Práctica dos: Sistema Cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

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Información general



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Asignatura: **Modelado de Sistemas Fisiológicos**

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Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('sistema2')
parameters.StopTime = '15'
```

```
parameters =
struct with fields:
    StopTime: '15'
```

```
parameters.Solver = 'ode15s'
```

```
parameters =
struct with fields:
    StopTime: '15'
      Solver: 'ode15s'
```

```
parameters.MaxStep = '1e-3'
```

```
parameters =
struct with fields:
    StopTime: '15'
      Solver: 'ode15s'
     MaxStep: '1e-3'
```

```
x=sim('sistema2',parameters)
```

```
x =
  Simulink.SimulationOutput:

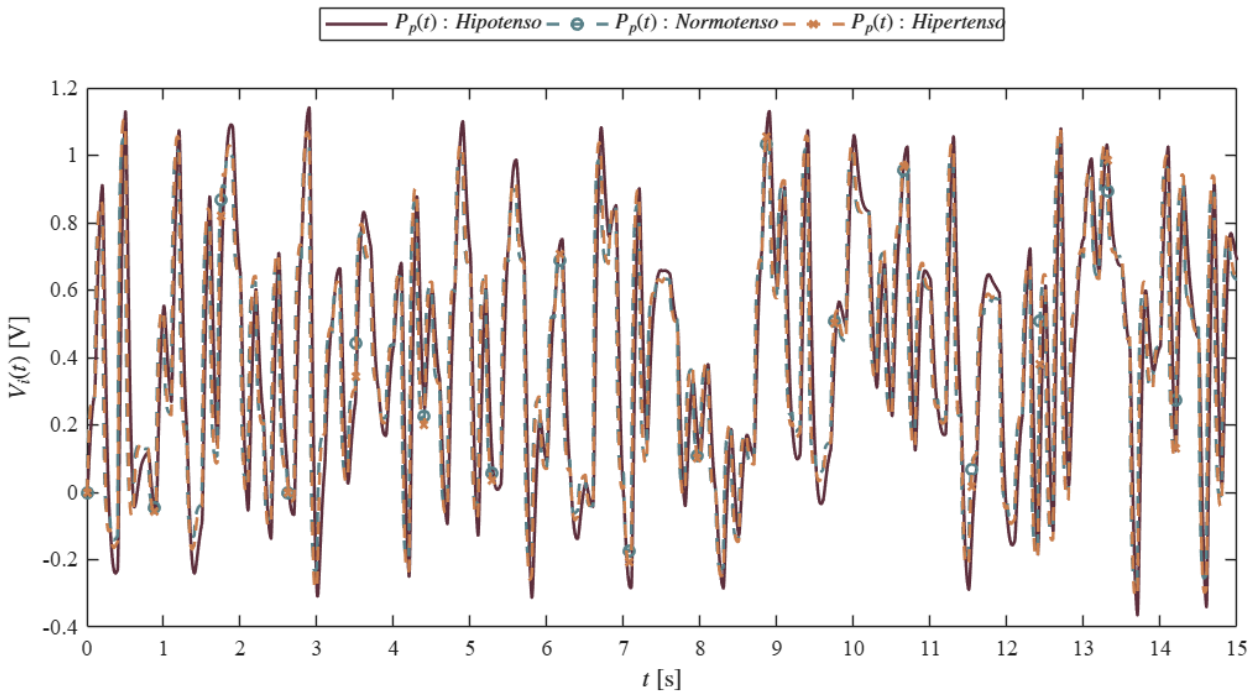
                  PIDL: [16895x1 double]
                 PIDLH: [16895x1 double]
                    Pa: [151x1 double]
                   PpH: [16895x1 double]
                   PpL: [16895x1 double]
                   PpN: [16895x1 double]
                     t: [16895x1 double]

     SimulationMetadata: [1x1 Simulink.SimulationMetadata]
           ErrorMessage: [0x0 char]
```

```
t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data, x.Pa, t, 'previous');
writematrix(Pa,'signal.xlsx')
% set_param('sistema/Vs(t): CE/Vs','VectorFormat','1-D array');
```

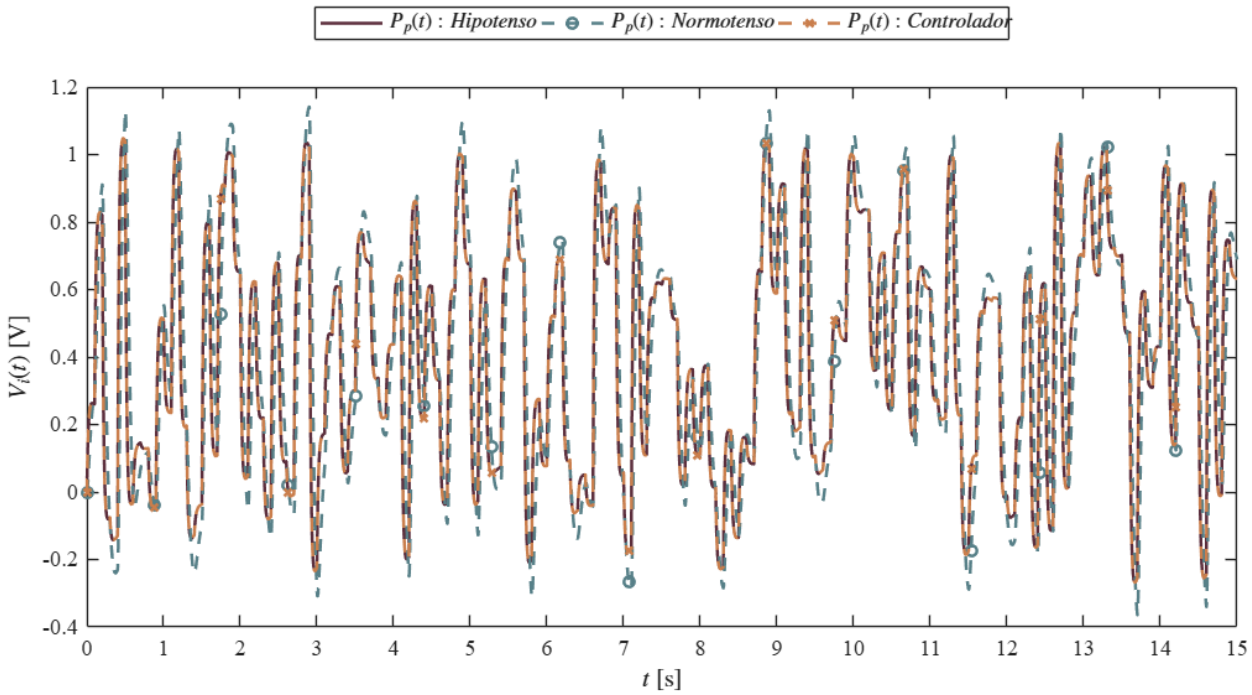
Respuestas: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t,x.PpL,x.PpN,x.PpH,'Individuos')
```



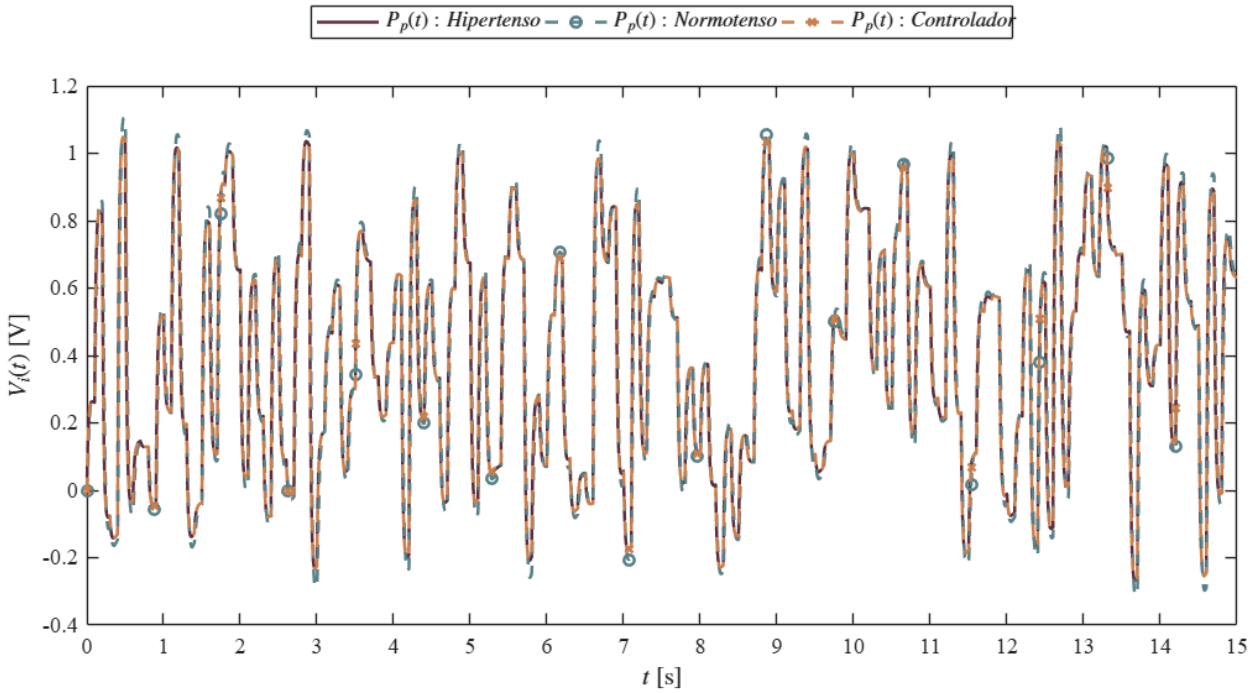
Respuestas: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'Hipotenso')
```



Respuestas Normotenso, Hipertenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpH,x.PIDLH,'Hipertenso')
```



Modelo de EDOs

```
par_hipo = [0.010,0.001,1.000,3.500]; %[Z,L,R,C]
par_normo = [0.010,0.001,1.000,1.500];
par_hiper = [0.010,0.001,1.000,1.000];
PpN = cardio(par_normo,Pa,t)
```

```
PpN = 15001×1
         0
    0.0161
    0.0313
    0.0457
    0.0592
    0.0719
    0.0839
    0.0952
    0.1059
    0.1159
    0.1253
    0.1342
    0.1426
    0.1505
    0.1579
      ⋮
```

```
PpL = cardio(par_hipo,PpN,t)
```

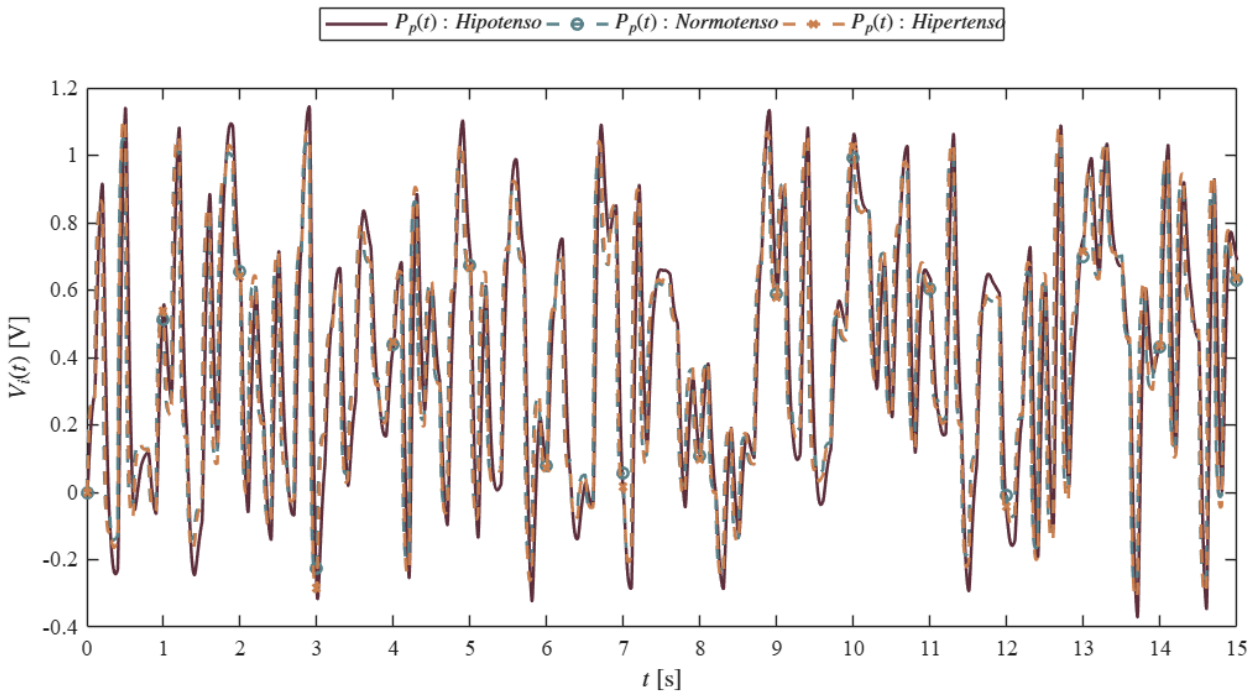
```
PpL = 15001×1
         0
         0
    0.0005
    0.0013
    0.0026
    0.0043
```

```
    0.0062
    0.0085
    0.0111
    0.0139
    0.0169
    0.0202
    0.0236
    0.0273
    0.0310
      ⋮
```

```
PpH = cardio(par_hiper,PpN,t)
```

```
PpH = 15001×1
         0
         0
    0.0016
    0.0046
    0.0088
    0.0139
    0.0198
    0.0264
    0.0335
    0.0410
    0.0489
    0.0570
    0.0652
    0.0735
    0.0818
      ⋮
```

```
plotsignals(t,PpL,PpN,PpH,'Individuos')
```



```
function Pp = cardio(par,Pa,t)
    Z = par(1); L = par(2); R = par(3); C = par(4);
    dt = t(2);
    n = round(t(end)/dt);

    FL = zeros(length(t),1); FL(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;

    for i = 1:n
        FL(i+1) = FL(i) + (-(1/L)*Pp(i) + (1/L)*Pa(i))*dt;
        Pp(i+1) = Pp(i) + ((1/C)*FL(i) - (1/(Z*C) + 1/(R*C))*Pp(i) +1/
(Z*C)*Pa(i))*dt;
    end
end
```

Funcion: Respuestas de la simulacion

```
function plotsignals(t,PpN,PpL,PpH,signal)
    set(figure(),'color','w')
    set(gcf,'units','inches','position', [1,1,10,5])
    set(gca, 'fontname','times new roman','fontsize',11)
    hold on; grid off; box on;
    clr1 = [92, 46, 60]/255;
    clr2 = [87, 128, 136]/255;
    clr3 = [202, 126, 76]/255;
    order = [clr1;clr2;clr3]; colororder(order)

    plot(t,PpN,'-',t,PpL,'--o',t,PpH,'--
x','LineWidth',1.5,'MarkerSize',5,'MarkerIndices',...
        1:1e3:length(t));


    xlabel('$t$ [s]','Interpreter','latex')
    ylabel('$V_i(t)$ [V]', 'Interpreter','latex')

    xlim([0 15]); xticks(0:1:15)

    if signal == "Individuos"
        ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
        L = legend('$P_p(t): Hipotenso$','$P_p(t): Normotenso$','$P_p(t):
Hipertenso$');

set(L,'interpreter','latex','location','northoutside','orientation','horizontal')
    elseif signal == "Hipotenso"
        ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
        L = legend('$P_p(t):Hipotenso$','$P_p(t): Normotenso$','$P_p(t):
Controlador$');

set(L,'interpreter','latex','location','northoutside','orientation','horizontal')
```

```
    elseif signal == "Hipertenso"
        ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)
        L = legend('$P_p(t):Hipertenso$','$P_p(t):Normotenso$','$P_p(t):
Controlador$');

set(L,'interpreter','latex','location','northoutside','orientation','horizontal')
    end

    exportgraphics(gcf,[signal, '_matlab.pdf'],'contenttype','vector')

end
```